

KING SHAKA INTL, South AfricaWMO: **685920**Lat: **29.602S**Lon: **31.124E**Elev: **358**StdP: **14.51**Time Zone: **2.00 (E02)**Period: **10-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	44.9	47.9	33.5	28.5	60.7	37.0	32.7	58.4	23.2	64.6	21.4	64.9	2.7	210	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.6	86.4	72.6	84.4	72.7	82.6	72.2	76.2	82.6	75.2	81.5	74.3	80.3	12.1	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.8	127.6	79.7	73.2	125.2	79.2	71.9	119.6	78.0	39.8	82.8	38.9	81.9	38.0	80.3	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.2	20.9	18.9	40.0	95.8	1.3	3.5	39.0	98.3	38.3	100.4	37.6	102.4	36.6	104.9		
(5)			37.9	78.8	1.4	1.1	36.9	79.6	36.1	80.3	35.4	80.9	34.4	81.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	68.3	74.7	74.8	73.9	69.4	66.3	62.0	60.7	63.2	65.7	67.2	69.9	72.7	(6)
(7)		DBStd	5.94	2.88	2.85	3.04	3.79	3.36	3.37	3.05	4.25	4.09	3.94	3.99	3.73	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	418	0	0	0	5	25	100	137	86	37	23	4	1	(9)
(10)		CDD50	6697	764	696	742	582	507	361	332	409	472	533	596	703	(10)
(11)		CDD65	1639	299	276	277	137	66	11	4	30	59	91	150	239	(11)
(12)		CDH74	8308	1692	1608	1489	641	354	163	115	229	234	282	495	1006	(12)
(13)		CDH80	1266	247	238	207	90	48	34	20	50	55	41	100	136	(13)
(14)	Wind	WSAvg	8.2	9.2	8.7	8.1	7.4	6.4	6.2	6.7	7.9	8.8	9.5	9.6	9.4	(14)
(15)	Precipitation	PrecAvg	33.8	4.7	3.5	3.4	2.0	1.3	0.8	1.7	1.3	2.3	3.9	3.9	4.9	(15)
(16)		PrecMax	45.4	11.9	9.0	8.1	5.4	5.7	2.7	7.6	4.3	7.9	10.7	10.4	12.6	(16)
(17)		PrecMin	16.6	1.9	0.6	1.1	0.4	0.0	0.0	0.0	0.1	0.5	0.8	1.3	1.6	(17)
(18)		PrecStd	7.8	2.2	2.3	2.0	1.3	1.4	0.8	2.4	1.0	1.5	2.1	2.0	2.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.7	86.7	87.3	86.4	85.7	84.3	82.8	85.9	85.8	85.7	88.4	87.6	(19)
(20)			MCWB	76.3	74.9	74.1	71.4	64.5	60.8	59.6	62.5	64.6	70.9	73.3	74.4	(20)
(21)		2%	DB	84.4	84.5	84.2	82.3	80.4	77.9	77.0	79.5	79.5	79.2	82.5	83.2	(21)
(22)			MCWB	74.7	74.6	74.0	71.8	67.1	60.2	60.2	63.5	67.4	69.0	72.4	74.0	(22)
(23)		5%	DB	82.5	82.6	82.3	79.2	77.3	75.0	73.5	75.5	75.6	76.9	78.7	80.7	(23)
(24)			MCWB	73.7	73.6	73.0	70.0	66.7	61.1	60.7	63.3	66.4	68.3	70.3	72.5	(24)
(25)		10%	DB	80.8	80.9	80.6	77.4	75.3	72.3	71.3	72.7	73.5	74.8	76.6	78.9	(25)
(26)			MCWB	72.4	72.4	72.4	69.3	66.4	61.1	61.0	62.9	65.4	67.2	69.2	71.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.9	77.1	76.4	74.5	72.4	67.2	65.5	68.4	70.6	72.1	75.1	77.2	(27)
(28)			MCDWB	84.4	84.1	82.3	82.4	79.2	73.2	71.9	76.4	77.9	80.2	83.2	84.3	(28)
(29)		2%	WB	76.1	75.7	75.3	72.8	69.9	65.2	64.2	66.7	69.0	70.5	72.9	75.2	(29)
(30)			MCDWB	82.2	82.2	81.3	79.6	76.3	72.3	71.0	74.6	76.0	76.8	79.9	81.3	(30)
(31)		5%	WB	74.7	74.7	74.3	71.6	68.4	63.8	63.1	65.3	67.5	69.3	71.6	73.7	(31)
(32)			MCDWB	80.7	80.8	80.2	78.2	74.9	71.5	70.1	73.1	73.8	75.0	76.8	79.1	(32)
(33)		10%	WB	73.6	73.8	73.4	70.3	67.3	62.5	61.9	63.9	66.6	68.1	70.2	72.5	(33)
(34)			MCDWB	79.3	79.4	78.9	76.2	73.6	69.8	69.0	71.0	72.6	73.7	75.4	77.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.9	12.6	13.3	15.7	18.7	22.4	21.5	19.4	15.2	13.2	12.1	11.4	(35)
(36)			MCDBR	14.4	14.6	15.9	18.6	22.5	28.6	28.6	26.7	20.4	17.5	16.7	14.0	(36)
(37)			MCWBR	7.7	8.1	8.5	11.0	13.6	17.2	17.5	14.1	11.4	10.4	9.1	7.7	(37)
(38)		5% WB	MCDWB	13.2	13.8	14.7	17.1	19.4	23.3	23.3	21.3	18.4	16.1	14.9	13.6	(38)
(39)			MCWBR	7.8	8.1	8.4	10.5	12.8	15.4	15.8	12.6	11.0	10.2	8.8	7.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.450	0.438	0.431	0.402	0.376	0.360	0.370	0.432	0.481	0.494	0.465	0.458	(40)	
(41)		taud	2.312	2.349	2.349	2.392	2.430	2.450	2.410	2.225	2.109	2.118	2.235	2.273	(41)	
(42)		Ebn at Noon	284	282	272	264	256	253	255	250	253	263	278	283	(42)	
(43)		Edh at Noon	44	41	39	34	29	27	30	39	49	51	47	46	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1884	1850	1611	1307	1120	996	1062	1239	1427	1558	1717	1861	(44)	
(45)		RadStd	150	137	81	79	56	46	71	74	91	137	146	137	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page